

Regression Assignment

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement

Solution:

To predict the insurance charges

2.) Tell basic info about the dataset (Total number of rows, columns)

Solution:

The total number of rows and columns in dataset are 1338 rows x 6 columns

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Solution:

The pre-processing method is **pd.get_dummies ()** for converting Sex and smoker in insurance_pre dataset

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model

Solution:

Yes ,I develop many models with r2_score and push into git hub as regression assignment .

**5.) All the research values (r2_score of the models) should be documented.
(You can make tabulation or screenshot of the results.)**

Solution:

1) Simple Linear Regression

S.NO	Hyper Parameter	Value	R Score
1	fit_intercept	True	0.78
2	copy_X	True	0.78
3	positive	False	0.78

2. Support Vector Machine

S.NO	Hyper Parameter	Linear	RBF	Poly	Sigmoid
1	C10	-0.11032	-0.11080	-0.547	-1.7093
2	C100	0.5262	-0.11013	0.5440	1.7056
3	C500	0.6803	-0.6397	0.545	-1.786
4	C1000	0.6794	-0.6215	0.5219	-1.798
5	C2000	0.7013	-0.6860	0.5473	-1.5786
6	C3000	0.7312	-0.6809	0.5677	-2.0119

3. Decision Tree Regression

S.No	Criteion	Splitter	R_score
1	friedman	best	0.69
2	friedman	random	0.21
3	absolute error	best	0.68
4	absolute error	random	0.75
5	poisson	best	0.69
6	poisson	random	0.68
7	squared_root	best	0.67
8	squared_root	random	0.70
9	default	default	0.69

4. Random Forest

S.NO	Criterion	Max Features	N_Estimators	R Value
1	Mse	Auto	10	0.9716
2	Mse	Auto	50	0.9801
3	Mse	Sqrt	10	0.9271
4	Mse	Sqrt	100	0.97722
5	Mse	Log2	10	0.9271
6	Mse	Log2	100	0.9122
7	Mae	Auto	10	0.9265
8	Mae	Auto	100	0.972
9	Mae	Sqrt	10	0.9232
10	Mae	Sqrt	100	0.9764
11	Mae	Log2	10	0.9232
12	Mae	Log2	100	0.9764

6.) Mention your final model, justify why u have chosen the same

Solution:

The final model is using **Random forest algorithm** because it predict the **r2_score** accuracy is **0.97801**